Status	Finished
Started	Monday, 12 October 2026, 5:42 PM
Completed	Monday, 12 October 2026, 5:52 PM
Duration	9 mins 55 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O(1)
d. O👎 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Sequential search
d. Binary search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It works on unsorted lists
c. It always starts from the beginning of the list
d. It compares elements sequentially

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search continues in the left sublist
c. The list is sorted
d. The search ends successfully ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search continues until the end of the list
c. The search stops immediately ✓
d. The search backtracks to find duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Finding
c. Searching ✓
d. Mining

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By sorting the list first
d. By dividing the list into halves

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Compare the target element with the middle element in the list ✓
c. Sort the list
d. Divide the list into two equal parts

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. The list must be sorted
c. Number values should only be present
Number values should only be present
d. There should be direct access to the middle element in any sublist

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search does not require dividing the list
c. Binary search works on unsorted lists
d. Binary search can find multiple instances of the target element

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 99 ✓
b. 89 and 94
c. 90 and 94
d. 89 and 99

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- c. Must use a sorted array
- d. There must be a mechanism to access middle element directly

Question **13**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list is sorted
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Linear search ✓
- c. Merge search
- d. Bubble search

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues in the left or right sublist ✓
- c. The search continues from the beginning
- d. The search stops